

Example 2.7 Linear Column with Two Plastic Hinges

Consider that the SDOF column with two PHLs shown in Figure 2.9(a) is subjected to a lateral applied force F_o . Let the moment versus plastic rotation relationships of the two plastic hinges follow those plotted in Figure 2.9(b), where $m_{y1} = 2F_oL$, $m_{y2} = F_oL$, $K_{t1} = EI/L$, and $K_{t2} = 2EI/L$. These relationships can be summarized as:

$$\text{if } \begin{cases} m_1 \leq 2F_oL \\ m_1 > 2F_oL \end{cases}, \quad \text{then } \begin{cases} \theta_1'' = 0 \\ m_1 = 2F_oL + (EI/L) \theta_1'' \end{cases} \quad (2.77a)$$

$$\text{if } \begin{cases} m_2 \leq F_oL \\ m_2 > F_oL \end{cases}, \quad \text{then } \begin{cases} \theta_2'' = 0 \\ m_2 = F_oL + (2EI/L) \theta_2'' \end{cases} \quad (2.77b)$$

Based on the structure's configuration shown in Figure 2.9(a), the stiffness matrices are:

$$\mathbf{K} = \begin{bmatrix} 12EI/L^3 \end{bmatrix}, \quad \mathbf{K}' = \begin{bmatrix} 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L \end{bmatrix}, \quad \mathbf{K}'' = \begin{bmatrix} 4EI/L & 2EI/L \\ 2EI/L & 4EI/L \end{bmatrix} \quad (2.78)$$

and therefore,

$$\begin{bmatrix} 12EI/L^3 & 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 4EI/L \end{bmatrix} \begin{bmatrix} x \\ -\theta_1'' \\ -\theta_2'' \end{bmatrix} = \begin{bmatrix} F_o \\ m_1 \\ m_2 \end{bmatrix} \quad (2.79)$$

First assume that the structure is linear, this gives $\theta_1'' = \theta_2'' = 0$. Then using the first equation of Eq. (2.79) gives

$$\frac{12EI}{L^3} x = F_o \quad (2.80)$$

Solving for the displacement x in Eq. (2.80) gives

$$x = F_o L^3 / 12EI \quad (2.81)$$

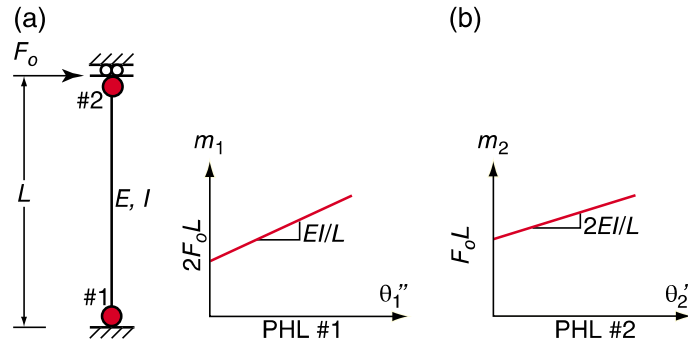


Figure 2.9 A SDOF system with two PHLs: (a) A single-degree-of-freedom system; (b) Moment versus plastic rotation relationships of the plastic hinges.

Then substituting Eq. (2.81) into the second and third equations of Eq. (2.79), the moments are calculated as

$$\begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{bmatrix} 6EI/L^2 \\ 6EI/L^2 \end{bmatrix} \left(\frac{F_o L^3}{12EI} \right) = \begin{Bmatrix} F_o L/2 \\ F_o L/2 \end{Bmatrix} \quad (2.82)$$

Comparing these moment demands in Eq. (2.82) with the yield moments in Eq. (2.77) shows that the plastic hinges have not yielded, which means the linear structure assumption is correct. Therefore, the final displacement and moments are presented in Eqs. (2.81) and (2.82), respectively.

Example 2.8 Nonlinear Column with Yielding in One Plastic Hinge

Consider again the SDOF column as shown in Figure 2.9(a) with the moment versus plastic rotation relationships shown in Figure 2.9(b). Now let the lateral applied force be equal to $3F_o$. The matrix equation used for solving the problem is similar to Eq. (2.79) except for the applied load, i.e.

$$\begin{bmatrix} 12EI/L^3 & 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 4EI/L \end{bmatrix} \begin{Bmatrix} x \\ -\theta_1'' \\ -\theta_2'' \end{Bmatrix} = \begin{Bmatrix} 3F_o \\ m_1 \\ m_2 \end{Bmatrix} \quad (2.83)$$

Again, the structure is first assumed to be linear, this gives $\theta_1'' = \theta_2'' = 0$. Then using the first equation of Eq. (2.83) gives

$$\frac{12EI}{L^3} x = 3F_o \quad (2.84)$$

Solving for the displacement x in Eq. (2.84) gives

$$x = F_o L^3 / 4EI \quad (2.85)$$

Then substituting Eq. (2.85) into the second and third equations of Eq. (2.83), the moments are calculated as

$$\begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{bmatrix} 6EI/L^2 \\ 6EI/L^2 \end{bmatrix} \left(\frac{F_o L^3}{4EI} \right) = \begin{Bmatrix} 3F_o L/2 \\ 3F_o L/2 \end{Bmatrix} \quad (2.86)$$

Comparing these moment demands in Eq. (2.86) with the yield moments in Eq. (2.77) shows that PHL #2 has yielded, but PHL #1 remains elastic. This means the original assumption that the structure is linear is incorrect.

Now assume that PHL #2 has yielded while PHL #1 remains elastic. This gives

$$\theta_1'' = 0, \quad m_2 = F_o L + (2EI/L) \theta_2'' \quad (2.87)$$

Substituting Eq. (2.87) into Eq. (2.83) and extracting the first and third equations of the matrix equation gives

$$\begin{bmatrix} 12EI/L^3 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L + 2EI/L \end{bmatrix} \begin{Bmatrix} x \\ -\theta_2'' \end{Bmatrix} = \begin{Bmatrix} 3F_o \\ F_o L \end{Bmatrix} \quad (2.88)$$

Solving for Eq. (2.88) gives

$$\begin{Bmatrix} x \\ -\theta_2'' \end{Bmatrix} = \begin{bmatrix} 12EI/L^3 & 6EI/L^2 \\ 6EI/L^2 & 6EI/L \end{bmatrix}^{-1} \begin{Bmatrix} 3F_o \\ F_o L \end{Bmatrix} = \begin{Bmatrix} F_o L^3 / 3EI \\ -F_o L^2 / 6EI \end{Bmatrix} \quad (2.89)$$

Then substituting Eq. (2.89) into the second and third equations of Eq. (2.83), the moments are calculated as

$$\begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{bmatrix} 6EI/L^2 & 2EI/L \\ 6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} F_o L^3 / 3EI \\ -F_o L^2 / 6EI \end{Bmatrix} = \begin{Bmatrix} 5F_o L / 3 \\ 4F_o L / 3 \end{Bmatrix} \quad (2.90)$$

Comparing the moment m_1 with the corresponding yield moment in Eq. (2.77) shows that PHL #1 remains elastic. This means the original assumption that PHL #2 has yielded and PHL #1 remains elastic is correct. Therefore, in summary,

$$x = \frac{F_o L^3}{3EI}, \quad \begin{Bmatrix} \theta_1'' \\ \theta_2'' \end{Bmatrix} = \begin{Bmatrix} 0 \\ F_o L^2 / 6EI \end{Bmatrix}, \quad \begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{Bmatrix} 5F_o L / 3 \\ 4F_o L / 3 \end{Bmatrix} \quad (2.91)$$

Example 2.9 Nonlinear Column with Yielding in All Plastic Hinges

Consider again the SDOF column as shown in Figure 2.9(a) with the moment versus plastic rotation relationships shown in Figure 2.9(b). Now let the lateral applied force be equal to $5F_o$. The matrix equation used for solving the problem is similar to Eq. (2.79) except for the applied load, i.e.

$$\begin{bmatrix} 12EI/L^3 & 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 4EI/L \end{bmatrix} \begin{Bmatrix} x \\ -\theta_1'' \\ -\theta_2'' \end{Bmatrix} = \begin{Bmatrix} 5F_o \\ m_1 \\ m_2 \end{Bmatrix} \quad (2.92)$$

Again, the structure is first assumed to be linear, this gives $\theta_1'' = \theta_2'' = 0$. Then using the first equation of Eq. (2.92) gives

$$\frac{12EI}{L^3} x = 5F_o \quad (2.93)$$

Solving for the displacement x in Eq. (2.93) gives

$$x = 5F_o L^3 / 12EI \quad (2.94)$$

Then substituting Eq. (2.94) into the second and third equations of Eq. (2.92), the moments are calculated as

$$\begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{bmatrix} 6EI/L^2 \\ 6EI/L^2 \end{bmatrix} \left(\frac{5F_o L^3}{12EI} \right) = \begin{Bmatrix} 5F_o L/2 \\ 5F_o L/2 \end{Bmatrix} \quad (2.95)$$

Comparing these moment demands in Eq. (2.95) with the yield moments in Eq. (2.77) shows that both PHLs #1 and #2 have yielded. This means the original assumption that the structure is linear is incorrect.

Now assume that both PHLs #1 and #2 have yielded. This gives

$$m_1 = 2F_o L + (EI/L) \theta_1'', \quad m_2 = F_o L + (2EI/L) \theta_2'' \quad (2.96)$$

Substituting Eq. (2.96) into Eq. (2.92) gives

$$\begin{bmatrix} 12EI/L^3 & 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 4EI/L + EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 4EI/L + 2EI/L \end{bmatrix} \begin{Bmatrix} x \\ -\theta_1'' \\ -\theta_2'' \end{Bmatrix} = \begin{Bmatrix} 5F_o \\ 2F_o L \\ F_o L \end{Bmatrix} \quad (2.97)$$

Solving for Eq. (2.97) gives

$$\begin{Bmatrix} x \\ -\theta_1'' \\ -\theta_2'' \end{Bmatrix} = \begin{bmatrix} 12EI/L^3 & 6EI/L^2 & 6EI/L^2 \\ 6EI/L^2 & 5EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 6EI/L \end{bmatrix}^{-1} \begin{Bmatrix} 5F_o \\ 2F_o L \\ F_o L \end{Bmatrix} = \begin{Bmatrix} 16F_o L^3/15EI \\ -3F_o L^2/5EI \\ -7F_o L^2/10EI \end{Bmatrix} \quad (2.98)$$

Then substituting Eq. (2.98) into the second and third equations of Eq. (2.92), the moments are calculated as

$$\begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{bmatrix} 6EI/L^2 & 4EI/L & 2EI/L \\ 6EI/L^2 & 2EI/L & 4EI/L \end{bmatrix} \begin{Bmatrix} 16F_o L^3/15EI \\ -3F_o L^2/5EI \\ -7F_o L^2/10EI \end{Bmatrix} = \begin{Bmatrix} 13F_o L/5 \\ 12F_o L/5 \end{Bmatrix} \quad (2.99)$$

These moments can similarly be obtained using Eq. (2.96), i.e.

$$m_1 = 2F_o L + \left(\frac{EI}{L} \right) \left(\frac{3F_o L^2}{5EI} \right) = \frac{13F_o L}{5} \quad (2.100a)$$

$$m_2 = F_o L + \left(\frac{2EI}{L} \right) \left(\frac{7F_o L^2}{10EI} \right) = \frac{12F_o L}{5} \quad (2.100b)$$

Comparing the moment demands m_1 and m_2 with the corresponding yield moments in Eq. (2.77) shows that both PHLs #1 and #2 have yielded. This means the original assumption is correct. Therefore, in summary,

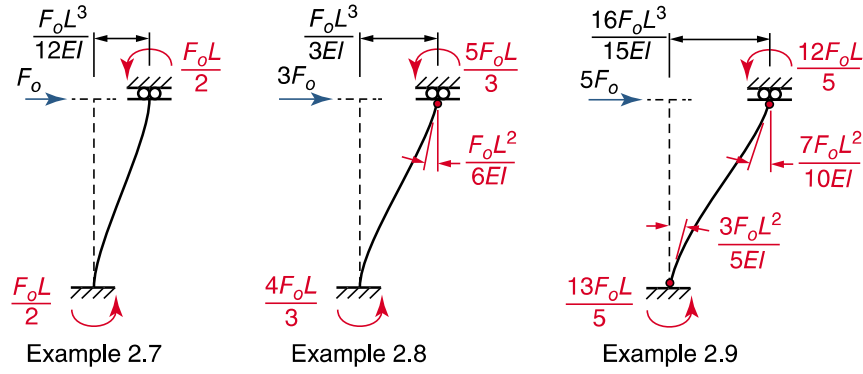


Figure 2.10 Graphical illustration of the responses of the SDOF system.

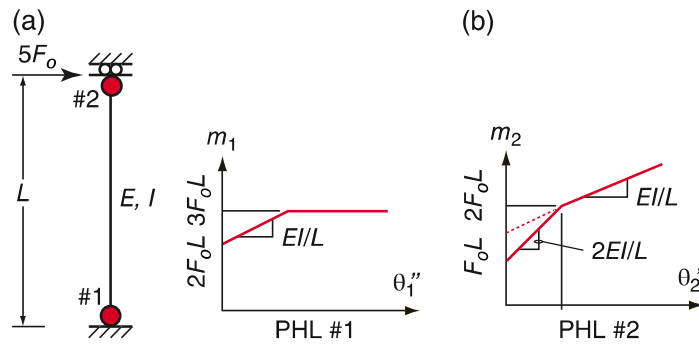


Figure 2.11 A SDOF system with two PHLs: (a) A single-degree-of-freedom system; (b) Moment versus plastic rotation relationships of the plastic hinges.

$$x = \frac{16F_o L^3}{15EI}, \quad \begin{Bmatrix} \theta_1'' \\ \theta_2'' \end{Bmatrix} = \begin{Bmatrix} 3F_o L^2 / 5EI \\ 7F_o L^2 / 10EI \end{Bmatrix}, \quad \begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{Bmatrix} 13F_o L / 5 \\ 12F_o L / 5 \end{Bmatrix} \quad (2.101)$$

Figure 2.10 shows a summary of the results in Examples 2.7, 2.8, and 2.9.

Example 2.10 Nonlinear Column with Tri-linear Plastic Rotation Behavior

Consider that the SDOF column with two PHLs shown in Figure 2.11(a) is subjected to a lateral applied force $5F_o$. Let the moment versus plastic rotation relationships of the plastic hinges follow the tri-linear relationships plotted in Figure 2.11(b). These relationships can be expressed in equation forms as:

$$\text{if } \begin{cases} m_1 \leq 2F_o L \\ 2F_o L < m_1 \leq 3F_o L \\ m_1 > 3F_o L \end{cases}, \quad \text{then } \begin{cases} \theta_1'' = 0 \\ m_1 = 2F_o L + (EI/L) \theta_1'' \\ m_1 = 3F_o L \end{cases} \quad (2.102a)$$

$$\text{if } \begin{cases} m_2 \leq F_o L \\ F_o L < m_2 \leq 2F_o L \\ m_2 > 2F_o L \end{cases}, \quad \text{then } \begin{cases} \theta_2'' = 0 \\ m_2 = F_o L + (2EI/L) \theta_2'' \\ m_2 = 3F_o L / 2 + (EI/L) \theta_2'' \end{cases} \quad (2.102b)$$